

Linear Statistical Models
BSMA3012

-Utkarsh Sahu (21F1001107)

(Q1)

Consider the Linear Model:

$$y_{ij} = \mu_i + \epsilon_{ij} ; 1 \leq j \leq n_i, 1 \leq i \leq 3$$

(a) If $\epsilon_{ij} \sim N(0,1)$, then find distribution of y_{ij} . Elaborate your Answer!

Solution:

Given $\epsilon_{ij} \sim N(0,1)$

$$\Rightarrow E(\epsilon_{ij}) = 0 \text{ \& } \text{Var}(\epsilon_{ij}) = 1$$

We know; if $E(x) = \mu$ \& $\text{Var}(x) = \sigma^2$
then $E(ax+b) = a\mu+b$ \& $\text{Var}(ax+b) = a^2\sigma^2$
if a \& b are constant.

Similarly!

$$E(y_{ij}) = E(\mu_i + \epsilon_{ij}) = E(\mu_i) + E(\epsilon_{ij})$$

Since μ_i is constant for 'i' therefore

$$E(\mu_i) = \mu_i \text{ \& we know } E(\epsilon_{ij}) = 0$$

therefore:

$$\boxed{E(y_{ij}) = \mu_i}$$

Now,

$$\text{Var}(y_{ij}) = \text{Var}(u_i + \epsilon_{ij})$$

Since u_i is constant therefore

$$\boxed{\text{Var}(y_{ij}) = \text{Var}(\epsilon_{ij}) = 1}$$

Therefore for:

$$y_{ij} = u_i + \epsilon_{ij}; 1 \leq j \leq n_i \text{ \& } 1 \leq i \leq 3$$

the distribution of y_{ij} must be:

$$\boxed{y_{ij} \sim N(u_i, 1)}$$

(Q2) Suppose $\underset{\sim}{Y} = \underset{\sim}{X}\underset{\sim}{\beta} + \underset{\sim}{\epsilon}$ be linear model,

$$\underset{\sim}{Y} = (y_1, y_2, \dots, y_n)^T,$$

$$\underset{\sim}{\beta} = (\beta_1, \beta_2, \dots, \beta_m)^T,$$

$$\underset{\sim}{X}_{ij} = ((x_{ij})) ; 1 \leq i \leq n \text{ \& } 1 \leq j \leq m$$

\&

$$\underset{\sim}{\epsilon} = (\epsilon_1, \epsilon_2, \dots, \epsilon_n)^T$$

$\underset{\sim}{\epsilon}$ has mean $\underset{\sim}{0}$ \& Var-Cov Matrix: $\sigma^2 \mathbf{I}_{n \times n}$

Let $\underset{\sim}{P}^T \underset{\sim}{\beta} = p_1 \beta_1 + p_2 \beta_2 + \dots + p_m \beta_m \quad \forall \underset{\sim}{\beta} \in \mathbb{R}^m$ then for

$\underset{\sim}{L} \in \mathbb{R}^n$ show that:

$$\mathbb{E}(\underset{\sim}{L}^T \underset{\sim}{Y}) = \underset{\sim}{P}^T \underset{\sim}{\beta}, \quad \forall \underset{\sim}{\beta} \in \mathbb{R}^m$$

Solution

We already know that for a linear model $\vec{y} = \mathbf{x}\vec{\beta} + \vec{e}$
 $E(\vec{y}) = \mathbf{x}\vec{\beta}$.

Now $\vec{l} \cdot \vec{y} = \vec{l}^T \vec{y} = (l_1, l_2, l_3, \dots, l_n) \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$
 $= l_1 y_1 + l_2 y_2 + \dots + l_n y_n$
 $= \sum_{i=1}^n l_i y_i$

Now, $E(\vec{l} \cdot \vec{y}) = E\left(\sum_{i=1}^n l_i y_i\right) = \sum_{i=1}^n (E(l_i y_i))$

[Since Expectation is a linear operator]

$$= \sum_{i=1}^n (l_i (E(y_i))) = l_1 E(y_1) + l_2 E(y_2) + \dots + l_n E(y_n)$$

$$= l_1 (\alpha_{11} \beta_1 + \alpha_{12} \beta_2 + \dots + \alpha_{1m} \beta_m) + \dots$$

$$+ l_n (\alpha_{n1} \beta_1 + \alpha_{n2} \beta_2 + \dots + \alpha_{nm} \beta_m)$$

$$= (l_1 \alpha_{11} + l_2 \alpha_{21} + l_3 \alpha_{31} + \dots + l_n \alpha_{n1}) \beta_1 + \dots$$

$$+ (l_1 \alpha_{1m} + l_2 \alpha_{2m} + \dots + l_n \alpha_{nm}) \beta_m$$

assuming $l_1 \alpha_{11} + l_2 \alpha_{21} + \dots + l_n \alpha_{n1} = p_1$

and $l_1 \alpha_{1m} + l_2 \alpha_{2m} + \dots + l_n \alpha_{nm} = p_m$

We have, $E(\vec{l}^T \vec{y}) = p_1 \beta_1 + p_2 \beta_2 + \dots + p_m \beta_m = \vec{p} \cdot \vec{\beta}$

$= \vec{p}^T \vec{\beta}$. Therefore, $E(\vec{l}^T \vec{y}) = \vec{p}^T \vec{\beta}; \forall \vec{\beta} \in \mathbb{R}^m$

```
# W3Q1b
```

```
"
```

Generate possible random values using R for μ_i and n_i

Note: The value of n_i 's should be greater than or equal to 100.

```
"
```

```
mu_values <- c(5, 10, 15) # mu_1=5, mu_2=10, mu_3=15
```

```
n_values <- sample(100:200, 3) # n_1, n_2, n_3, each >= 100
```

```
print("Generated mu values:")
```

```
print(mu_values)
```

```
print("Generated n values (n1, n2, n3):")
```

```
print(n_values)
```

```
n1 <- n_values[1]
```

```
n2 <- n_values[2]
```

```
n3 <- n_values[3]
```

```
n <- sum(n_values)
```

```
[1] "Generated mu values:"  
[1]  5 10 15  
[1] "Generated n values (n1, n2, n3):"  
[1] 115 150 139
```

```
# W3Q1c
```

```
"
```

Using R, generate the dataset, i.e. y_{ij} , and store it in a data frame.

```
"
```

```
y1 <- rnorm(n1, mean = mu_values[1], sd = 1)
```

```
y2 <- rnorm(n2, mean = mu_values[2], sd = 1)
```

```
y3 <- rnorm(n3, mean = mu_values[3], sd = 1)
```

```
# Create a data frame
```

```
dataset <- data.frame(  
  value = c(y1, y2, y3),
```

```
  group = factor(rep(1:3, times = n_values))  
)
```

```
print("the generated dataset:")
```

```
View(dataset)
```

dataset			
(row)	value	group	
1	6.9002	1	
2	3.3295	1	
3	6.3878	1	
4	7.2507	1	
5	4.1087	1	
6	6.8392	1	
7	5.847	1	
8	4.8081	1	
9	4.9878	1	
10	4.8469	1	
11	5.3802	1	

```
# W3Q1d
```

```
"
```

```
Using R, find the mean of yi's denoted by yio using the generated dataset.
```

```
"
```

```
y1_mean <- mean(y1)
```

```

y2_mean <- mean(y2)
y3_mean <- mean(y3)
group_means <- c(y1_mean, y2_mean, y3_mean)

```

```

print("Group means (y1_o, y2_o, y3_o):")
print(group_means)

```

```

print(group_means)
[1] "\nUsing R, find the mean of yi's denoted by yio using the generated dataset.\n"
[1] "Group means (y1_o, y2_o, y3_o):"
[1] 5.001499 9.964252 15.060723

```

W3Q1e

"

Using R, compute the following: [3 Marks]

- i. Sum of squares within group, i.e. SSW.
- ii. Sum of squares between group, i.e. SSB.
- iii. Total sum of squares, i.e. T SS.

"

i

```

grand_mean <- mean(dataset$value)
ssw1 <- sum((y1 - y1_mean)^2)
ssw2 <- sum((y2 - y2_mean)^2)
ssw3 <- sum((y3 - y3_mean)^2)
SSW <- ssw1 + ssw2 + ssw3
print(paste("SSW:", SSW))

```

ii

```

SSB <- n1 * (y1_mean - grand_mean)^2 +
      n2 * (y2_mean - grand_mean)^2 +
      n3 * (y3_mean - grand_mean)^2
print(paste("SSB:", SSB))

```

iii

```

TSS <- sum((dataset$value - grand_mean)^2)
print(paste("TSS:", TSS))

```

```

print(paste("TSS:", TSS))
[1] "\nUsing R, compute the following: [3 Marks]\ni. Sum of squares within group, i.e. SSW.\nii. Sum of squares between group, i.e. SSB.\niii. Total sum of squares, i.e. T SS.\n"
[1] "SSW: 371.120367579996"
[1] "SSB: 6395.78409179142"
[1] "TSS: 6766.90445937142"

```

r\$>

```
# W3Q1f
```

```
"
```

Using R, verify that $TSS = SSB + SSW$.

```
"
```

```
print(paste("SSB + SSW:", SSB + SSW))
```

```
print(paste("TSS:", TSS))
```

```
print(paste("Is TSS equal to SSB + SSW?", isTRUE(all.equal(TSS, SSB + SSW))))
```

```
Using R, verify that TSS = SSB + SSW.
"
print(paste("SSB + SSW:", SSB + SSW))
print(paste("TSS:", TSS))
print(paste("Is TSS equal to SSB + SSW?", isTRUE(all.equal(TSS, SSB + SSW))))
[1] "\nUsing R, verify that TSS = SSB + SSW.\n"
[1] "SSB + SSW: 6766.90445937142"
[1] "TSS: 6766.90445937142"
[1] "Is TSS equal to SSB + SSW? TRUE"
```

```
# W3Q1g
```

```
"
```

Randomly generate a number between 1000 and 1500 by using command `rdunif()` and perform that many iterations for the parts (c),(d) and (e) in R.

```
"
```

```
if (!requireNamespace("extraDistr", quietly = TRUE)) install.packages("extraDistr")
```

```
library(extraDistr)
```

```
set.seed(123)
```

```
num_iterations <- rdunif(1, min = 1000, max = 1500)
```

```
print(paste("Number of iterations to perform:", num_iterations))
```

```
ssw_values <- numeric(num_iterations)
```

```
ssb_values <- numeric(num_iterations)
```

```
tss_values <- numeric(num_iterations)
```

```
# Loop for iterations
```

```
for (k in 1:num_iterations) {
```

```
  # (c) Generate dataset
```

```
  y1 <- rnorm(n1, mean = mu_values[1], sd = 1)
```

```
  y2 <- rnorm(n2, mean = mu_values[2], sd = 1)
```

```
  y3 <- rnorm(n3, mean = mu_values[3], sd = 1)
```

```
  all_y <- c(y1, y2, y3)
```

```
  # (d) Calculate means
```

```
  y1_mean <- mean(y1)
```

```
  y2_mean <- mean(y2)
```

```
  y3_mean <- mean(y3)
```

```

grand_mean <- mean(all_y)

# (e) Compute SSW, SSB, TSS
SSW <- sum((y1 - y1_mean)^2) + sum((y2 - y2_mean)^2) + sum((y3 - y3_mean)^2)
SSB <- n1 * (y1_mean - grand_mean)^2 + n2 * (y2_mean - grand_mean)^2 + n3 *
(y3_mean - grand_mean)^2
TSS <- sum((all_y - grand_mean)^2)

# Store results
ssw_values[k] <- SSW
ssb_values[k] <- SSB
tss_values[k] <- TSS
}

# W3Q1h

"Using R, store the values of SSW, SSB and TSS computed in part (g) in a data
frame.
"

# Create the results data frame
results_df <- data.frame(
  Iteration = 1:num_iterations,
  SSW = ssw_values,
  SSB = ssb_values,
  TSS = tss_values
)

print("The simulation results:")
View(results_df)

```

results_df					
(row)	Iteration	SSW	SSB	TSS	
1	1	418.7739	6238.9929	6657.7667	
2	2	387.1293	6437.8106	6824.94	
3	3	430.6999	6250.6906	6681.3904	
4	4	385.0634	6242.6805	6627.7439	
5	5	417.1507	6406.5994	6823.7502	
6	6	414.5434	6287.7997	6702.3431	
7	7	359.4416	6515.0106	6874.4522	
8	8	380.8566	6385.751	6766.6076	
9	9	384.9014	6171.4636	6556.365	
10	10	386.7255	6079.9401	6466.6656	
11	11	354.9795	6263.003	6617.9825	

```

p_ssw <- ggplot(results_df, aes(x = SSW)) +
  geom_histogram(aes(y = ..density..), bins = 30, fill = "lightblue", color = "black") +
  ggtitle("Histogram of Sum of Squares Within Groups (SSW)")

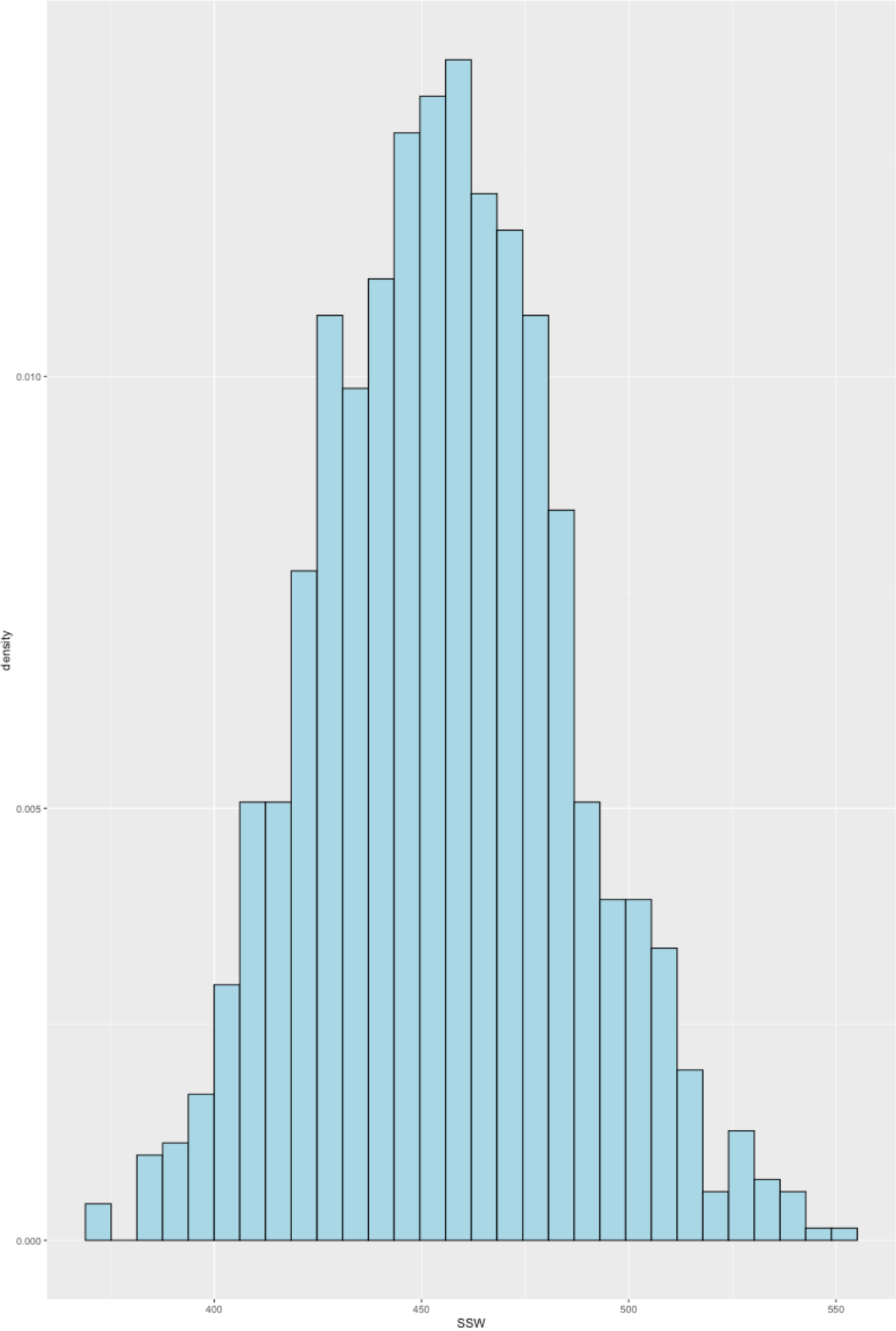
# Histogram for SSB
p_ssb <- ggplot(results_df, aes(x = SSB)) +
  geom_histogram(aes(y = ..density..), bins = 30, fill = "lightgreen", color = "black") +
  ggtitle("Histogram of Sum of Squares Between Groups (SSB)")

# Histogram for TSS
p_tss <- ggplot(results_df, aes(x = TSS)) +
  geom_histogram(aes(y = ..density..), bins = 30, fill = "salmon", color = "black") +
  ggtitle("Histogram of Total Sum of Squares (TSS)")

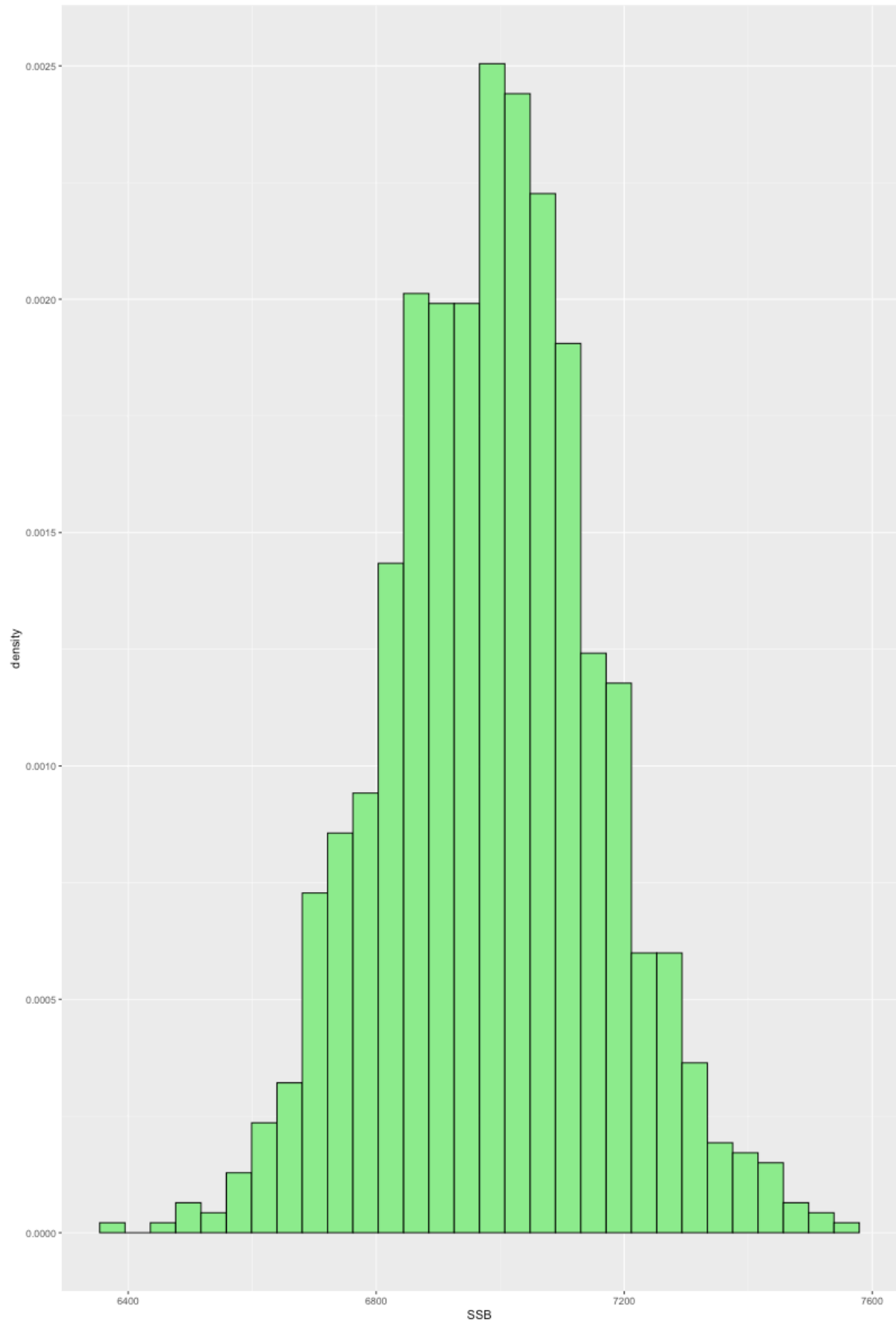
# Print the plots
print(p_ssw)
print(p_ssb)
print(p_tss)

```

Histogram of Sum of Squares Within Groups (SSW)



Histogram of Sum of Squares Between Groups (SSB)



Histogram of Total Sum of Squares (TSS)

